DAY 12

```python
y = iris.target

# Create Random Forest model

rf = RandomForestClassifier(

    n_estimators=100,   # number of trees

    oob_score=True,     # enable Out-of-Bag error

    random_state=42

)

# Train model

rf.fit(X, y)

# 1. OOB Score (Out-of-Bag Accuracy)

print("OOB Accuracy:", rf.oob_score_)

print("OOB Error:", 1 - rf.oob_score_)

# 2. Feature Importance

importance = pd.DataFrame({

    "Feature": iris.feature_names,

    "Importance": rf.feature_importances_

})

importance = importance.sort_values(by="Importance", ascending=False)

print("\nFeature Importance:")

print(importance)

# 3. Individual Prediction Example

sample = np.array([[5.1, 3.5, 1.4, 0.2]])

print("\nPrediction:", rf.predict(sample))
```

```python
[1]: from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier
from sklearn import tree
#Load dataset
iris=load_iris()
x=iris.data
y=iris.target

#Create Decision Tree model

dt=DecisionTreeClassifier(
    criterion="gini", #can also use entropy
    max_depth=3     #pruning(pre-pruning)
)
#train model
dt.fit(x,y)

#prdiction
print(dt.predict([[5.1,3.5,1.4,0.2]]))
```

[0]

```python
[2]: from sklearn.datasets import load_iris

from sklearn.ensemble import RandomForestClassifier

import pandas as pd

import numpy as np

# Load dataset

iris = load_iris()

X = iris.data

y = iris.target

# Create Random Forest model

rf = RandomForestClassifier(
```

```
OOB Accuracy: 0.9533333333333334
OOB Error: 0.046666666666666634

Feature Importance:
             Feature  Importance
2  petal length (cm)    0.436130
3   petal width (cm)    0.436065
0  sepal length (cm)    0.106128
1   sepal width (cm)    0.021678

Prediction: [0]
```

[3]:

```python
# 1. IMPORT LIBRARIES

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.svm import SVC

from sklearn.metrics import (

    accuracy_score,

    precision_score,

    recall_score,

    f1_score,

    confusion_matrix,

    roc_auc_score

)

import numpy as np

# 2. LOAD DATASET

iris = load_iris()

X = iris.data
```

Code JupyterLab Python [conda env:base] * Anaconda Toolb

```python
y = iris.target

# 3. TRAIN-TEST SPLIT

X_train, X_test, y_train, y_test = train_test_split(

    X, y,

    test_size=0.2,

    random_state=42

)

# 4. CREATE SVM MODEL

# kernel options: 'linear', 'rbf', 'poly', 'sigmoid'

svm = SVC(

    kernel='rbf',       # non-linear kernel

    probability=True,  # needed for ROC-AUC

    random_state=42

)

# 5. TRAIN MODEL

svm.fit(X_train, y_train)

# 6. PREDICTION

y_pred = svm.predict(X_test)

# probability scores (for ROC-AUC)

y_prob = svm.predict_proba(X_test)

# 7. EVALUATION METRICS

print("MODEL PERFORMANCE")
```

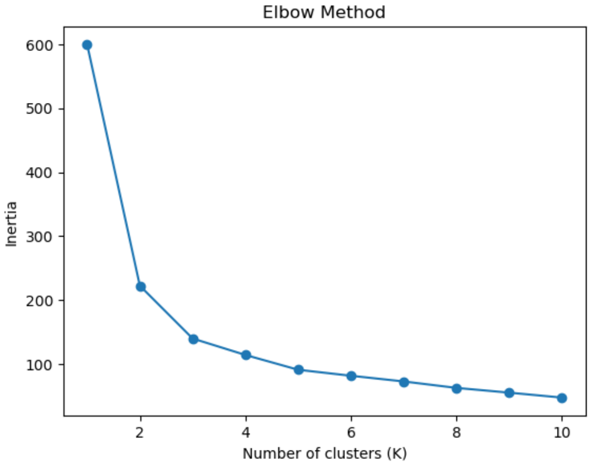
```python
print("MODEL PERFORMANCE")

# Accuracy

acc = accuracy_score(y_test, y_pred)

print("Accuracy:", acc)

# Precision

precision = precision_score(y_test, y_pred, average='macro')

print("Precision:", precision)

# Recall

recall = recall_score(y_test, y_pred, average='macro')

print("Recall:", recall)

# F1 Score

f1 = f1_score(y_test, y_pred, average='macro')

print("F1 Score:", f1)

# Confusion Matrix

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:\n", cm)

# ROC-AUC (multi-class)

roc = roc_auc_score(y_test, y_prob, multi_class='ovr') #One-vs-Rest approach

print("\nROC-AUC Score:", roc)

# 8. SINGLE PREDICTION

sample = np.array([[5.1, 3.5, 1.4, 0.2]])

prediction = svm.predict(sample)
```

Code JupyterLab Python [conda env:base] * Anaconda Too

```
MODEL PERFORMANCE
Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1 Score: 1.0

Confusion Matrix:
 [[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

ROC-AUC Score: 1.0

Single Sample Prediction: [0]
```

[4]:

```python
import warnings

warnings.filterwarnings("ignore")

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from sklearn.datasets import load_iris

from sklearn.cluster import KMeans

from sklearn.metrics import silhouette_score

from sklearn.preprocessing import StandardScaler

# 1. Load dataset

iris = load_iris()

X = iris.data

# 2. Feature scaling (important for K-Means)

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)
```

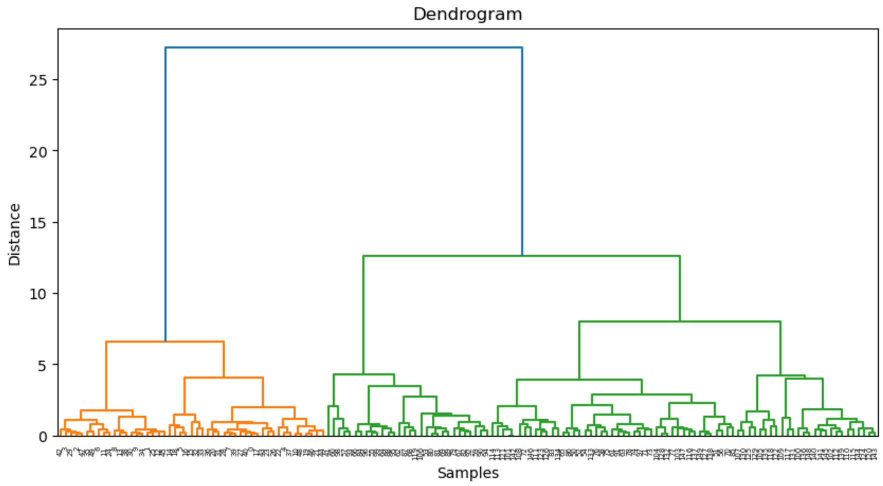



```
Silhouette Score: 0.45994823920518635
```


```
# 3. Elbow Method (Inertia)

inertia = []

K_range = range(1, 11)

for k in K_range:

    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)

    kmeans.fit(X_scaled)

    inertia.append(kmeans.inertia_)

plt.plot(K_range, inertia, marker='o')

plt.title("Elbow Method")

plt.xlabel("Number of clusters (K)")

plt.ylabel("Inertia")

plt.show()

# 4. Fit final KMeans (choose K = 3 for iris)

kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)

labels = kmeans.fit_predict(X_scaled)

# 5. Silhouette Score

score = silhouette_score(X_scaled, labels)

print("Silhouette Score:", score)

# 6. Visualize clusters (first 2 features)

plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=labels, cmap='viridis')

plt.scatter(kmeans.cluster_centers_[:, 0],

            kmeans.cluster_centers_[:, 1],
```

Code JupyterLab Python [conda env:base] * Anaconda

```python
    n_clusters=3,

    linkage='ward'

)

labels = hc.fit_predict(X_scaled)

print("Cluster Labels:")

print(labels)
```




```
Cluster Labels:
[1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 2 1 1 1 1 1 1 1 1 0 0 0 2 0 2 0 2 0 2 2 0 2 0 2 0 2 2 2 2 0 0 0 0
 0 0 0 0 0 2 2 2 2 0 2 0 0 2 2 2 2 0 2 2 2 2 2 0 2 2 0 0 0 0 0 0 2 0 0 0 0
 0 0 0 0 0 0 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0]
```

[5]:

```python
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris

from sklearn.preprocessing import StandardScaler

from scipy.cluster.hierarchy import dendrogram, linkage

from sklearn.cluster import AgglomerativeClustering

# Load dataset

iris = load_iris()

X = iris.data

# Scale features

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

# Create linkage matrix

linked = linkage(X_scaled, method='ward')

# Plot dendrogram

plt.figure(figsize=(10,5))

dendrogram(linked)

plt.title("Dendrogram")

plt.xlabel("Samples")

plt.ylabel("Distance")

plt.show()

# Hierarchical Clustering
```